GE ENERGY POWER CONVERSION
RUGBY, ENGLAND

Ref: TYRS4
Design No: 49259/69.5.25
Frame Size: A220VV4
Date: 25 Sep 2024

PROJECT: RS4 STANDARD DESIGN – 69MWS

NAMEPLATE RATING

Rated kVAr	pf	Poles	Phases	Voltage (V)	Current (A)	Frequency (Hz)	Speed (Rev/Min)
42500	0	4	3	15000	1636	50	1500

Overload capability: 50 MVAr over-excited (Class F)

Enclosure: IP54 IC8A1W7 (CACW Totally enclosed water cooled)
Stator winding connection: STAR
Governing Standard: IEC 60034

Maximum stator operating temperature	120	deg C by resistance
Maximum stator operating temperature	122	deg C by ETD
Maximum field operating temperature	130	deg C by resistance
Design water inlet temperature	40	deg C
Maximum operating altitude	1000	metres
Insulation system	Class	F
Total temperatures	Class	B
Overspeed	1800	rev/min

PERFORMANCE OF SALIENT POLE MACHINE

All performance figures are subject to tolerance where defined in IEC 60034.

- Total loss at nameplate rating: +15% as per Table 22 of IEC60034-1 Ed 14, 2022.
- Direct axis transient and sub-transient reactances: +/-15% as per clause 4.20 of IEC60034-3, 2020.
- Average A-weighted sound pressure level: +3dBA as per Table 22 of IEC60034-1 Ed 14, 2022.
- Moment of inertia is a guaranteed minimum value with no -ve tolerance.

All losses, pu resistances and time constants are quoted at a temperature of 95 deg C.

Input kVAr	42500	21250	34000	17000	No Load
Stator current amp	1636	818	1309	654	0
Rotor current amp	538	381	37	137	241
Fixed loss kW	373	373	373	373	373
Total loss	670	475	494	410	395
Power factor	0	0	0	0	
compared with 1 pf	over-excited	over-excited	under-excited	under-excited	

Stored Energy : 69 MWs (based on the guaranteed minimum value of inertia)

Instantaneous Short Circuit Capacity (Nominal) : 607 MVA
(Based on nominal value of saturated sub-transient reactance at time = 0 s)

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Base impedance 5.294 ohms

			Rated I (unsaturated)	Rated V (saturated)	
Synchronous reactance	D-Axis	Xd	1.103		pu
Transient reactance	D-Axis	X'd	0.144	0.115	pu
Sub-transient reactance	D-Axis	X"d	0.097	0.070	pu
Synchronous reactance	Q-Axis	Xq	0.589		pu
Sub-transient reactance	Q-Axis	X"q	0.131	0.078	pu
Stator winding leakage reactance		Xl	0.047		pu
Negative phase sequence reactance		X2	0.098 pu		
Potier reactance		Xpot	0.122 pu at rated load		
Stator winding dc resistance		Ra	0.00239		pu
Positive phase sequence resistance		R1	0.00413		pu
Negative phase sequence resistance		R2	0.021		pu
Zero phase sequence reactance		X0	0.036		pu
Zero phase sequence resistance		R0	0.0144		pu
OC transient time constant DA		T'do	8.74		s
SC transient time constant DA		T'd	1.142	0.814	s
OC sub-transient time constant DA		T"do	0.038		s
SC sub-transient time constant DA		T"d	0.026	0.047	s
OC sub-transient time constant QA		T"qo	0.142		s
SC sub-transient time constant QA		T"q	0.032		s
Armature dc time constant		Ta	0.169	0.111	s

Stator winding dc resistance (phase)	0.00976	ohm at 20 C
Field winding dc resistance	0.25	ohm at 20 C
Exciter armature dc resistance (phase)	0.00616	ohm at 20 C
Exciter field winding dc resistance	9.57	ohm at 20 C
Permanent field protective resistor	27	ohm

Short circuit ratio	1.015	
Maximum kVAr available at 0 pf under-excited	0.8	pu
Maximum kVAr available at 0 pf over-excited	1	pu
Saturation factor S(1.0)	1.12	
Saturation factor S(1.2)	1.83	

EXCITATION

	Main field current	Main field voltage	Exciter field current	Exciter field voltage
Rated load at zero pf over-excited	538	188	6.5	76
Rated voltage on open circuit	240	84	2.9	34
Rated current on short circuit	237	83	2.9	34

Exciter response from rated load IEC 60034 3.21 pu/s

MECHANICAL

Rotor Inertia WR2	5595	kg.m2
Rotor inertia constant	1.624	s
Cooling water quantity (machine + lube oil unit)	20.5	L/s
Expected mean sound pressure level @ 1m (ISO1680)	85	dBA

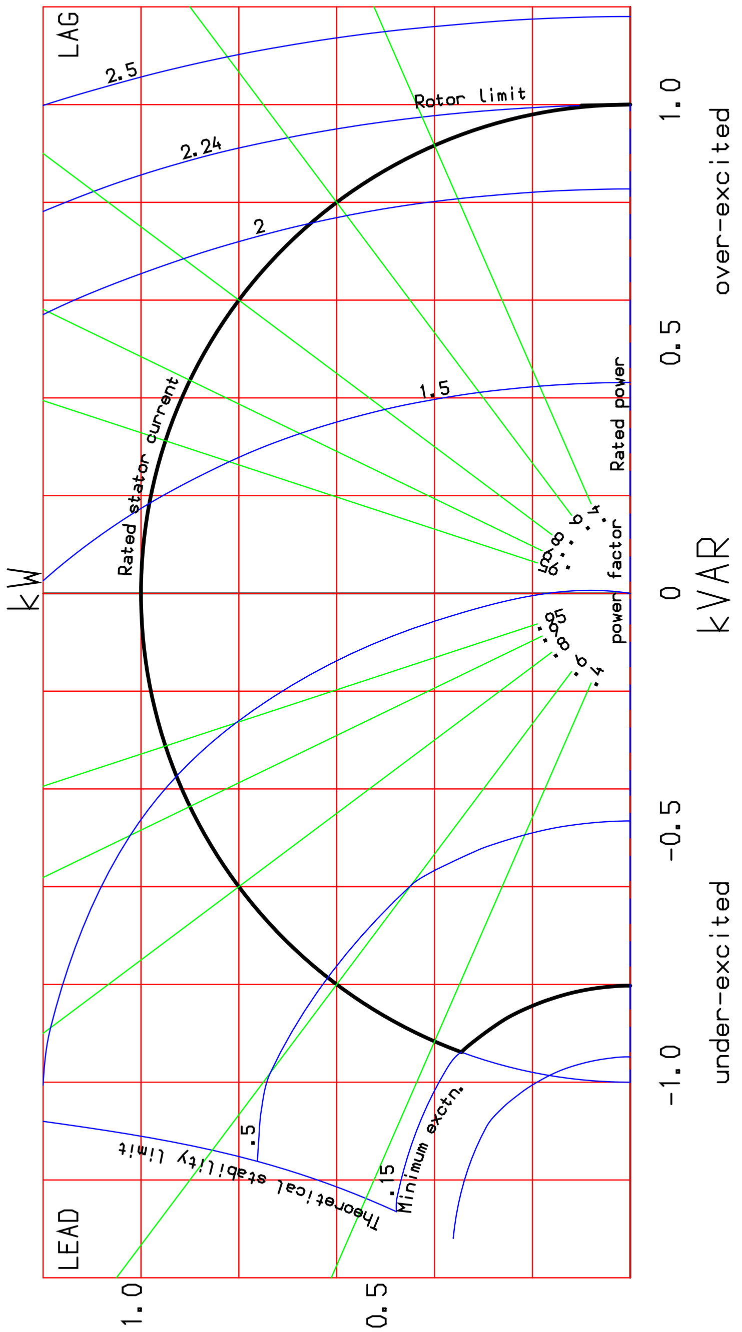
GE Energy Power Conversion

Curve: 49259/69/PC
Ref.: TYRS4

SYNCHRONOUS MACHINE POWER CHART

RS4 Standard Design

Synchronous Generator 42500kVA 0PF 4 Poles 15kV 50Hz 1500 rev/min
1 pu kVA = 42500   1 pu Field Current = 240.4 Amps



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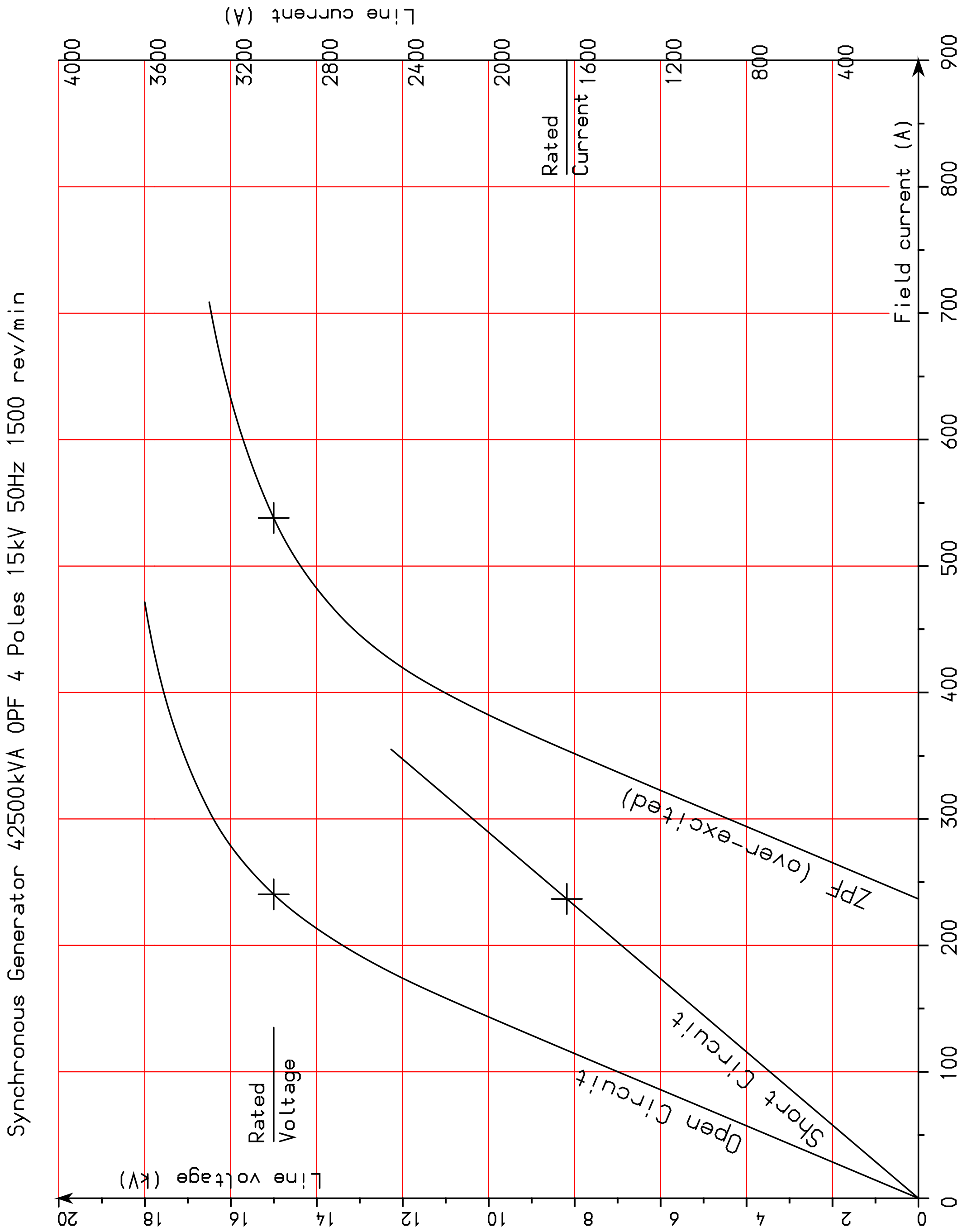
GE Energy Power Conversion

Curve: 49259/69/SAT

Ref. TYRS4

SYNCHRONOUS MACHINE SATURATION CURVES

RS4 Standard Design



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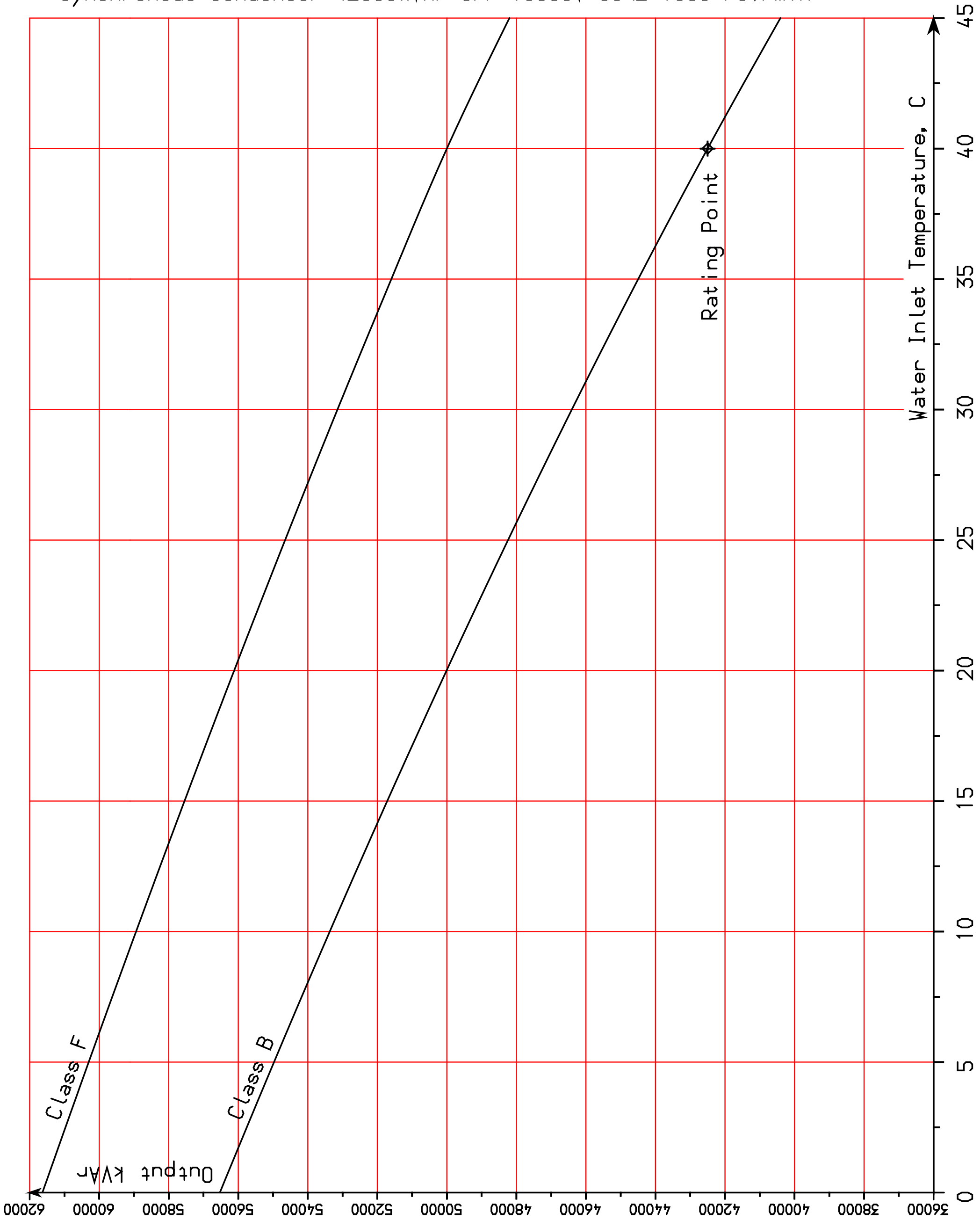
GE Energy Power Conversion

Curve: RS4 Std/GT
Ref. TYRS4

RS4 CAPABILITY CURVE

RS4 Standard Design
Synchronous Condenser 42500kVAr 0PF 15000V 50Hz 1500 rev/min



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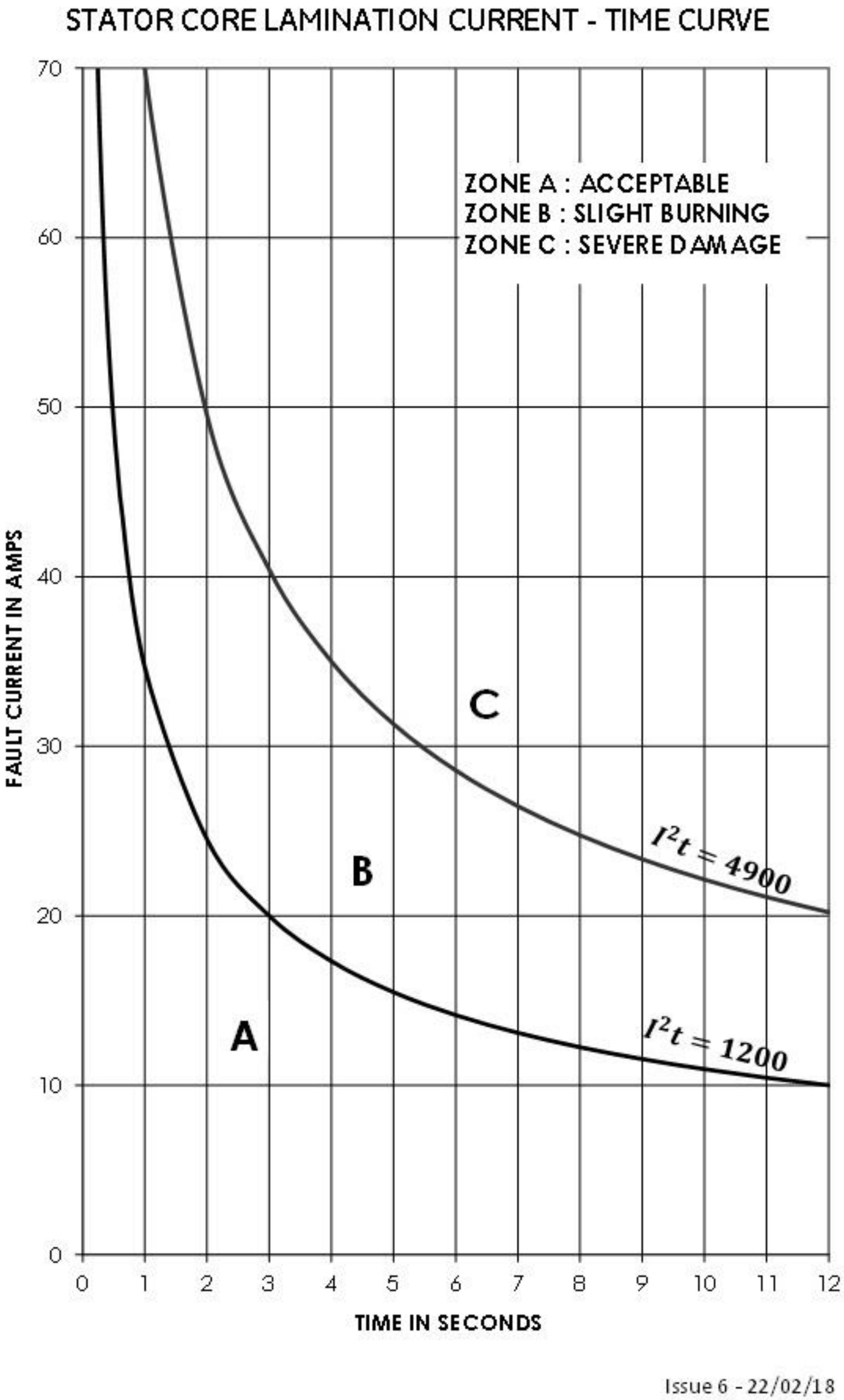
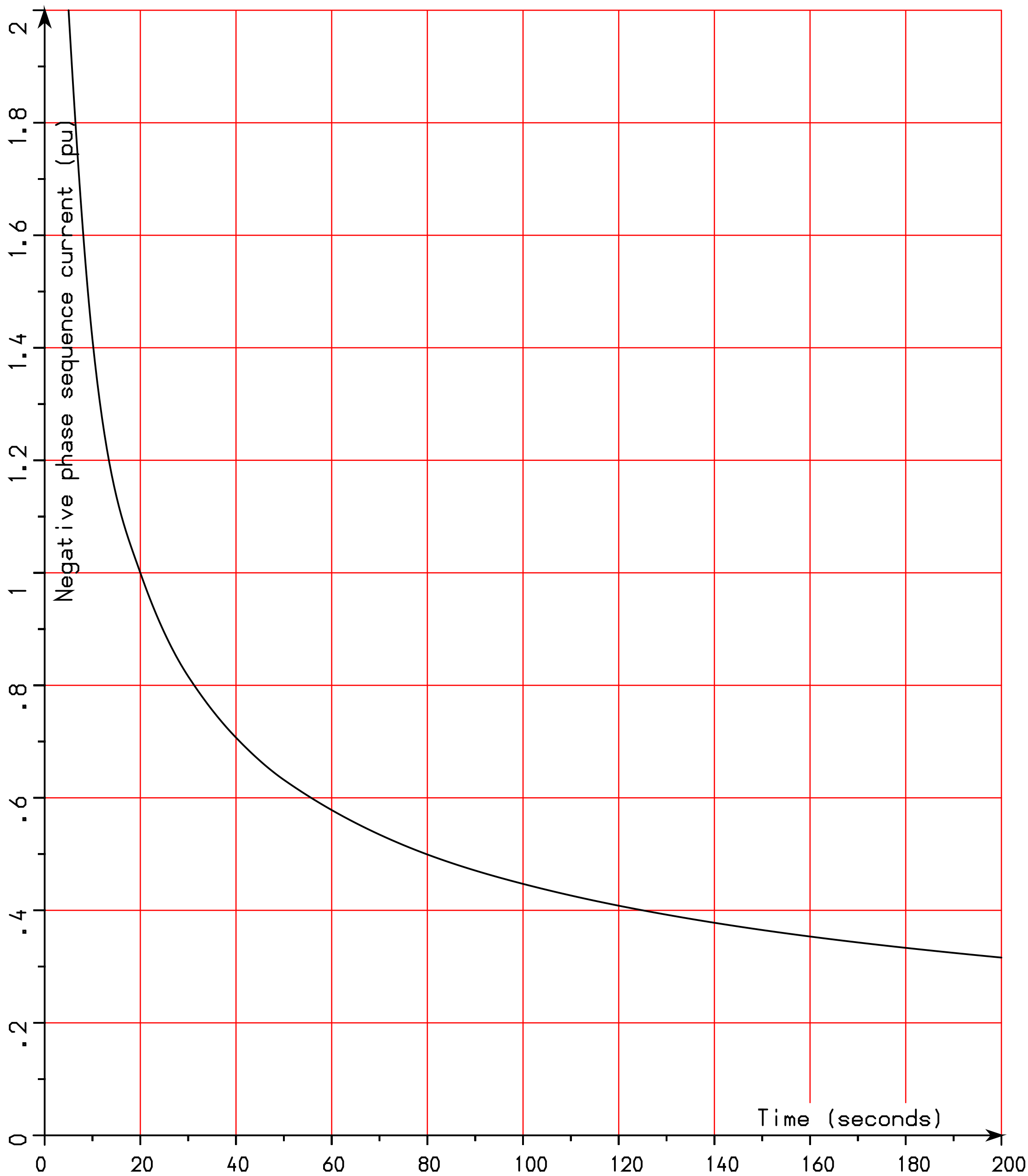

STATOR CORE LAMINATION CURRENT - TIME CURVE
ZONE A : ACCEPTABLE
ZONE B : SLIGHT BURNING
ZONE C : SEVERE DAMAGE
A
B
C
$I^2t = 4900$
$I^2t = 1200$
FAULT CURRENT IN AMPS
TIME IN SECONDS
0
10
20
30
40
50
60
70
0 1 2 3 4 5 6 7 8 9 10 11 12
Issue 6 - 22/02/18

GE Energy Power Conversion

GE Energy Power Conv Ref.: NSEQ

NEGATIVE PHASE SEQUENCE CURRENT WITHSTAND

Maximum continuous negaitive sequence current is 0.10pu
for synchronous compensators as defined in Table 2 of
BS EN60034-1 (2010) & IEC60034-1 (2022)
The curve below plots $I_2^2t = 20$ for short term conditions



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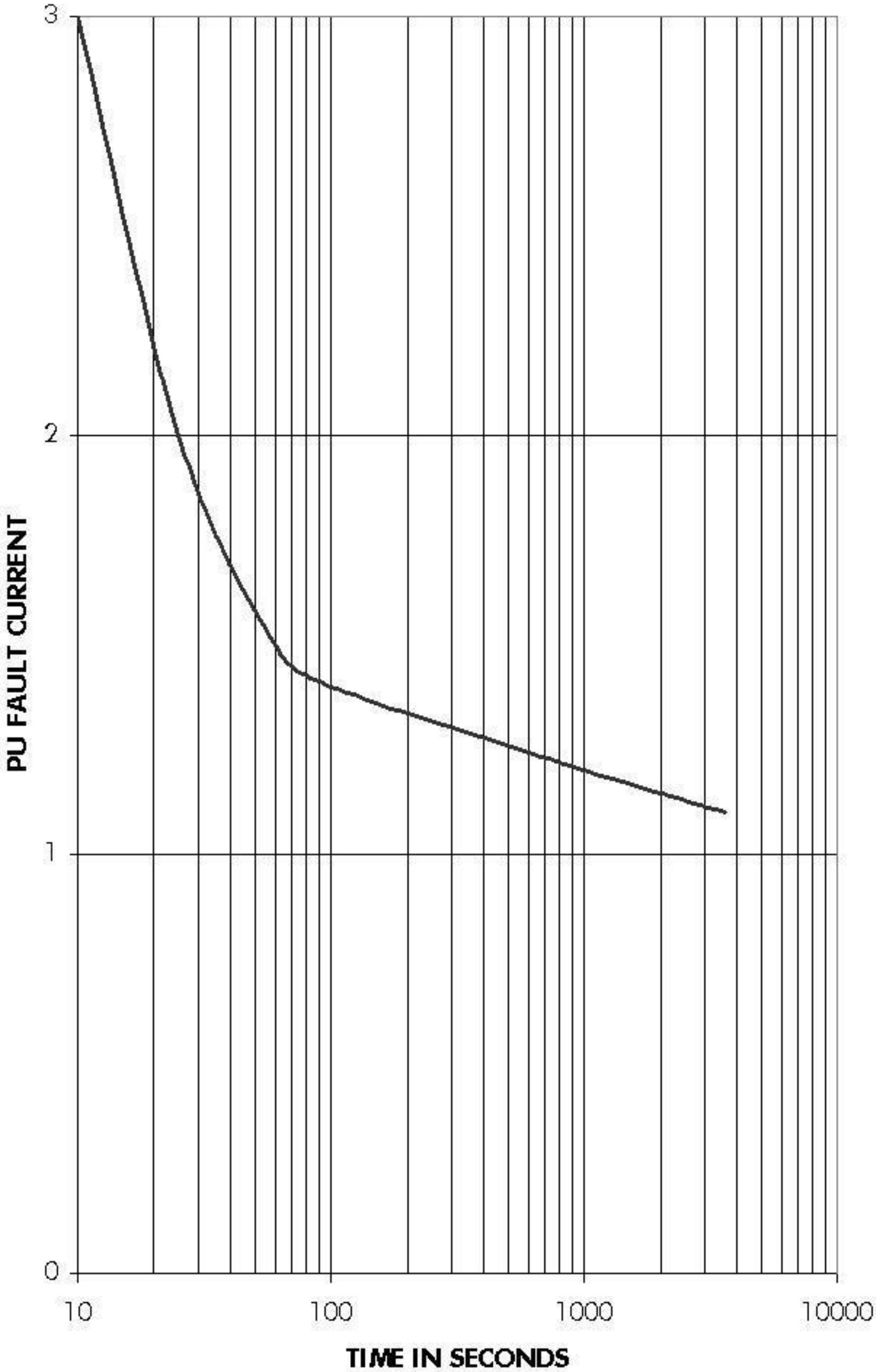
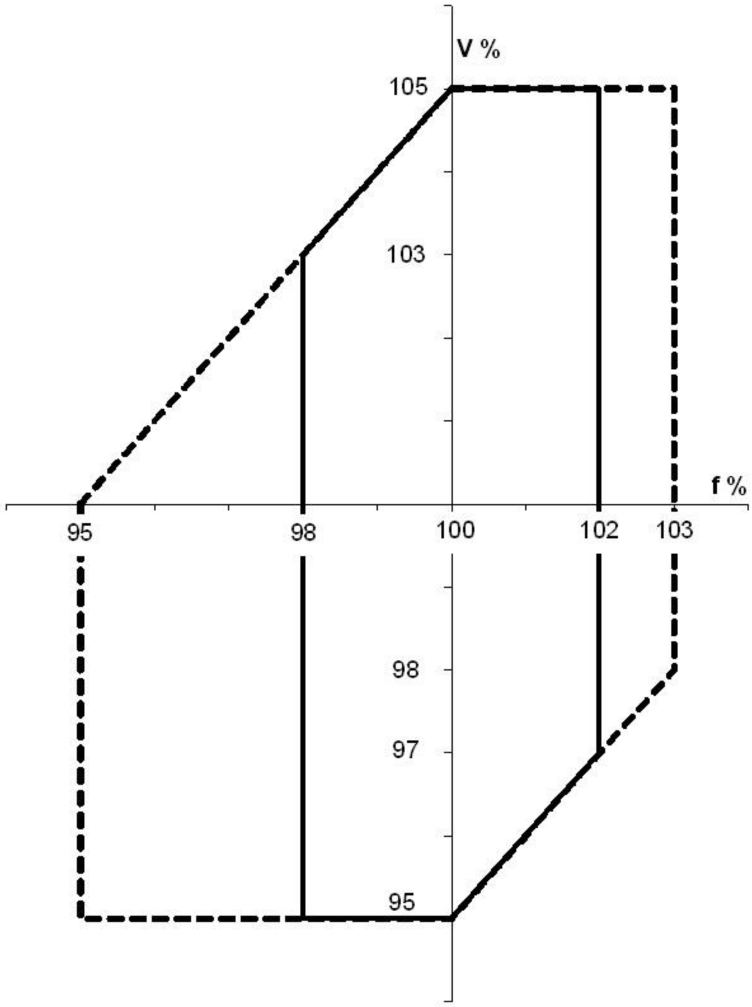
THERMAL DAMAGE CURVE



Issue 5 - 07/03/13

Operation Over Ranges of Voltage and Frequency



In accordance with clause 4.6 of BS EN IEC 60034-3:2020

Synchronous generators and compensators shall be capable of continuous rated output at the rated power factor over the ranges of ±5% in voltage and ±2% in frequency, as defined by the solid boundary above.

The temperature rise limits in the applicable tables of clause 8.10 of IEC 60034-1 & BS EN 60034-1 shall apply at the rated voltage and frequency.

Continuous operation at rated output at certain parts of the solid boundary may cause additional temperature rise of up to approximately 10K.

Generators or compensators will also carry output at rated power factor within the ranges of ±5% in voltage and -5% to +3% in frequency, as defined by the dotted boundary above, but temperature rises will be further increased. Therefore, operation outside the solid boundary should be limited in extent, duration and frequency of occurrence in order to minimise the reduction of lifetime due to thermal effects. The output should be reduced or other corrective measures taken as soon as practicable.

Potentially required operation outside the dotted boundary above shall be the subject of an agreement.

Issue 9 - 11/03/2022